

Interdisciplinary Undergraduate Project

Course Code: UGQ301

Fruit Juice Classification Using Impedance and Phase Measurement Circuit

*A project report submitted in partial fulfilment of the requirements
for the Interdisciplinary Undergraduate Project (UGQ301)*

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Abstract

This report presents an interdisciplinary study on the non-destructive classification of commercial fruit juices using **Electrical Impedance Spectroscopy (EIS)** integrated with machine learning. The project encompasses the design of an analogue impedance measurement circuit, data-driven analysis, and the development of a deployable classification device.

In contrast to wide-spectrum analysis, this system operates at a **single fixed frequency of 758.58 Hz**, chosen for its stability in capturing the electrochemical signatures of different liquid compositions. Impedance magnitude and phase measurements were collected for various fruit juice varieties from the same brand, including **Mango, Mosambi (Sweet Lime), and Orange**. The core data processing involved training a **K-Nearest Neighbour (KNN)** classifier on features refined via **Linear Discriminant Analysis (LDA)**.

By establishing a robust transfer function to map raw voltage ratios to physical impedance, the model successfully distinguishes between juice types based on their unique interfacial and bulk electrical properties. This streamlined, single-frequency approach demonstrates that high-accuracy classification can be achieved with significantly reduced hardware complexity. The resulting system provides a foundation for a compact, real-time juice identification device suitable for rapid quality screening in the food and beverage industry.

Keywords: Electrical Impedance , Fruit Juice Classification, Machine Learning, KNN, LDA, Single-Frequency Analysis, Embedded Systems.

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Chapter 1

Introduction

1.1 Background and Motivation

Food quality and authenticity are critical concerns in the modern food industry. Fruit juices, being among the most consumed beverages worldwide, are frequent targets of adulteration—a practice involving the addition of cheaper ingredients such as water, sugar, artificial flavours, or other low-grade juices to increase volume and profit margins. Traditional analytical methods for quality assessment, such as high-performance liquid chromatography (HPLC), nuclear magnetic resonance (NMR) spectroscopy, and gas chromatography–mass spectrometry (GC-MS), are highly accurate but require specialised laboratory equipment, trained personnel, and significant time, making them impractical for rapid or on-site testing.

Electrical impedance spectroscopy (EIS) offers a promising low-cost alternative. By measuring the electrical impedance—comprising resistance and reactance—of a liquid sample across a range of frequencies, EIS captures the electrochemical and physical properties of the sample. These properties are characteristic of the sample’s composition, making EIS a powerful tool for non-destructive quality assessment. When combined with modern machine learning algorithms, EIS-derived features can be leveraged to build robust, automated classifiers capable of identifying juice type and detecting adulteration with high accuracy.

This project explores the complete pipeline from EIS data collection to a deployable, hardware-integrated classification device—pursued as a collaborative interdisciplinary effort between the electrical engineering and data science teams.

1.2 Problem Statement

The central challenge addressed in this project is:

- Can electrical impedance measurements, taken at multiple frequencies, reliably distinguish between different types of fruit juices?
- Which subset of measurement frequencies provides sufficient discriminative power for accurate classification while minimising hardware complexity?

- How can machine learning models be integrated with a low-cost hardware device for practical, real-world use?

Practically, measuring impedance at 50 different frequencies in a portable device is computationally expensive and hardware-intensive. The key objective of the DSAI phase is therefore to identify a minimal set of 2–3 optimal frequencies that retain maximum classification accuracy, thereby simplifying the hardware requirements for the final device.

1.3 Objectives

The specific objectives of this project are as follows:

1. Design and implement an electrical impedance measurement circuit capable of measuring impedance magnitude and phase across a range of frequencies.
2. Collect, clean, and preprocess a multi-frequency impedance dataset for multiple juice samples.
3. Perform exploratory data analysis and visualise class separability using LDA projections.
4. Develop and evaluate a machine learning classification pipeline using LDA dimensionality reduction and KNN classification.
5. Identify the optimal 2–3 measurement frequencies through systematic brute-force combinatorial search.
6. Validate the reduced-frequency model and compare its performance with the full-spectrum baseline.
7. Provide actionable frequency recommendations to the hardware team for circuit calibration.

1.4 Scope of the Report

This report covers all three phases of the project. Chapter 2 describes the hardware phase (Phase 1), covering the design of the impedance measurement circuit. Chapter 3 presents the DSAI phase (Phase 2) in detail, including data preprocessing, exploratory analysis, model development, and optimal frequency selection. Chapter 4 outlines the integration phase (Phase 3) and future work. Conclusions are presented in Chapter 5.

The GitHub repository containing all source code is available at:

https://github.com/YOUR_GITHUB_LINK_HERE

Chapter 2

Phase 1: Hardware Design of the Impedance Measurement Circuit

2.1 Overview

implementing signal generation, magnitude measurement, and phase detection blocks on a unified circuit,

2.2 Circuit Design Principles

DC Power Generation Block: ICL7660 Implementation

The DC Power Generation block utilizes the **ICL7660** monolithic CMOS voltage converter to provide the dual-rail supply necessary for the system’s precision operational amplifiers. The design is based on the **Switched-Capacitor (Charge Pump)** principle:

- **Charge Phase:** An internal oscillator (Pin 7) controls a network of MOS switches that charge the *Flying Capacitor* (connected between Pins 2 and 4) to the level of the input voltage ($V_{in} = +12V$) [?].
- **Transfer Phase:** The switches reconfigure to connect the positive terminal of the flying capacitor to ground, effectively “flipping” the voltage. This charge is transferred to the *Reservoir (Output) Capacitor* at Pin 5 [?].
- **Result:** This process generates a negative output voltage at Pin 5, calculated as:

$$V_{OUT} \approx -V^+ \tag{2.1}$$

where $V_{OUT} = -12V$, creating the stable negative rail required for the bipolar operation of the LF411 op-amps

Magnitude Measurement Block: Design Principles

The design principle for this stage is **Transimpedance Amplification and Precision Rectification**.

- **Current-to-Voltage Conversion:** The impedance being measured is placed in the feedback path of **OP2**, which acts as a transimpedance-style amplifier where the output voltage is directly proportional to the total impedance magnitude.
- **Reference Branching:** A parallel “reference line” uses **OP1** with fixed resistors to provide a stable baseline signal, enabling the calculation of a ratio ($\frac{V_{sig}}{V_{ref}}$) that is independent of excitation voltage fluctuations.
- **Envelope Detection:** Diodes **D1** and **D2**, combined with 10nF capacitors and 10k Ω resistors, function as peak detectors to convert AC sinusoidal signals into stable DC voltages (V_1 and V_2) for ADC sampling.
- **Dynamic Range Scaling:** Range Changing Switches (**S2**, **S3**, **S23**) allow for gain adjustment to accommodate varying conductivity levels across different test samples.

Phase Detection Block: Design Principles

The design principle for this block is **Zero-Crossing Detection and Pulse-Width Modulation (PWM) Conversion**.

- **Squaring Circuits:** Op-amps **OP3** and **OP4** are configured as high-gain comparators that convert sinusoidal “signal” and “reference” waveforms into square waves by triggering at zero-crossing points.
- **XOR Phase Comparison:** A **74HC86N** XOR gate compares these square waves, producing a pulse string where the duty cycle is directly proportional to the phase shift (θ) between the two signals.

Chapter 3

Phase 2: Data Science and AI Analysis

3.1 Overview

Phase 2 constitutes the core DSAI contribution of this project. The objectives of this phase were to:

1. Preprocess and clean the raw impedance dataset.
2. Perform exploratory data analysis to understand class separability.
3. Develop a baseline machine learning classification pipeline.
4. Identify the optimal subset of 2–3 measurement frequencies.
5. Validate the reduced-frequency model.

3.2 Dataset Description

3.2.1 Raw Dataset: `juice_data.csv`

The raw dataset `juice_data.csv` contained impedance and phase measurements for multiple juice samples. Each record included:

Table 3.1: Structure of the raw dataset `juice_data.csv`

Field	Description
Juice Name	Categorical label identifying the type of fruit juice
Sample ID	Unique integer identifier for each sample
Timestamp	Time of measurement ($t = 0, 2, 4$ minutes)
$ Z _{f_1} - Z _{f_{50}}$	Impedance magnitude at 50 frequency points
$\phi_{f_1} - \phi_{f_{50}}$	Phase angle at 50 frequency points

The 50 measurement frequencies span the range from approximately 100 Hz to 1000 kHz. Three key frequencies identified later in this work are listed in Table 3.2.

Table 3.2: Key frequencies identified through brute-force selection

Column Index	Frequency (Hz)	Approximate Range
1	120.23	Low frequency
39	131,825.67	Mid-high frequency
42	229,086.77	High frequency

3.2.2 Data Characteristics

The dataset exhibited several structural irregularities in its raw form, including inconsistent column naming, mixed data types, missing values in certain frequency channels, and records from multiple timestamps requiring separation. The preprocessing steps described below addressed these issues systematically.

3.3 Data Preprocessing

3.3.1 Libraries Used

All preprocessing was performed in Python using the `pandas` and `numpy` libraries. The preprocessing pipeline involved the following steps.

3.3.2 Timestamp Selection

Measurements were available at three timestamps: $t = 0$, $t = 2$, and $t = 4$ minutes. A comparative analysis of the impedance spectra across timestamps revealed that readings at $t = 4$ exhibited the lowest intra-class variance and the highest signal stability, consistent with full electrode equilibration. Accordingly, **only the $t = 4$ readings were retained** for all downstream analysis.

3.3.3 Feature Construction

For each sample, the impedance magnitude and phase angle at all 50 frequencies were concatenated into a single feature vector of dimension 100:

$$\mathbf{x} = [|Z|_{f_1}, \phi_{f_1}, |Z|_{f_2}, \phi_{f_2}, \dots, |Z|_{f_{50}}, \phi_{f_{50}}] \in \mathbb{R}^{100} \quad (3.1)$$

This merged representation encodes the full spectral response of the sample and serves as input to all subsequent machine learning models.

3.4 Exploratory Data Analysis

3.4.1 LDA-Based Class Separability

Linear Discriminant Analysis (LDA) was applied to the full 100-dimensional feature matrix to project the data onto the two most discriminative dimensions. LDA maximises the Fisher criterion:

$$J(\mathbf{w}) = \frac{\mathbf{w}^\top S_B \mathbf{w}}{\mathbf{w}^\top S_W \mathbf{w}} \quad (3.2)$$

where S_B is the between-class scatter matrix and S_W is the within-class scatter matrix.

The resulting 2D LDA projection is shown in Figure 3.1. Each point represents one juice sample, coloured by juice class. The degree of class separation in the LDA space provides an upper bound on the classification accuracy achievable by a linear classifier on this dataset.

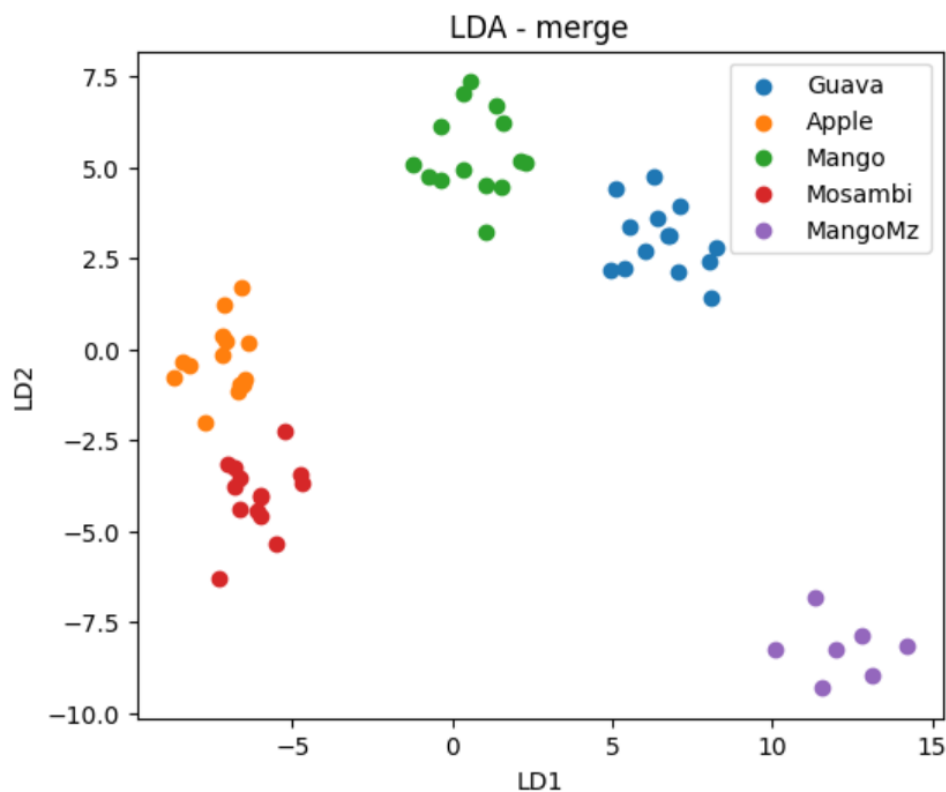


Figure 3.1: LDA 2D projection of the full impedance–phase feature space. Each colour represents a distinct juice class. Clear cluster separation is visible for most juice types.

The LDA plot demonstrates strong separability for several juice classes, with some overlap in specific class pairs attributable to similar ionic compositions. This motivates the use of LDA as a dimensionality reduction step in the classification pipeline.

3.5 Baseline Classification Pipeline: All 50 Frequencies

3.5.1 Train-Test Split

The dataset was split into training (80%) and test (20%) sets using stratified random sampling via `train_test_split` from `scikit-learn`, with a fixed random seed to ensure reproducibility. Stratification ensures that the class distribution is preserved in both splits, reducing the risk of evaluation bias.

3.5.2 LDA Dimensionality Reduction

LDA was fitted exclusively on the training set to prevent data leakage. The fitted LDA transform was then applied to both the training and test sets, reducing the 100-dimensional feature space to 2 dimensions.

$$\mathbf{X}_{\text{lda}} = \mathbf{X} \cdot \mathbf{W}_{\text{lda}} \quad (3.3)$$

where $\mathbf{W}_{\text{lda}} \in \mathbb{R}^{100 \times 2}$ contains the top two LDA discriminant vectors fitted on the training data.

3.5.3 KNN Classification with Hyperparameter Tuning

A K-Nearest Neighbour (KNN) classifier was trained on the 2D LDA-reduced training features. Hyperparameter optimisation was performed using `GridSearchCV` with 5-fold stratified cross-validation. The search grid was defined as:

Listing 3.1: KNN hyperparameter grid

```

1 param_grid = {
2     "n_neighbors": [3, 5, 7, 9, 11],
3     "weights":      ["uniform", "distance"],
4     "metric":       ["euclidean", "manhattan"]
5 }
```

Grid search parameters: `cv=5`, `scoring="accuracy"`, `n_jobs=-1`.

The best hyperparameter configuration identified by cross-validation was applied to train the final model, which was then evaluated on the held-out test set.

3.5.4 Baseline Results

The baseline KNN model trained on all 50 frequencies (100 features after merging impedance and phase) achieved:

Table 3.3: Baseline KNN classification results (all 50 frequencies)

Metric	Value
Test Accuracy	74.0%
Number of Features (pre-LDA)	100
LDA Output Dimensions	2
Cross-validation Folds	5

The confusion matrix and classification report for the baseline model are shown in Figure 3.2.

Test Accuracy: 0.7407407407407407				
Classification Report:				
	precision	recall	f1-score	support
Apple	0.83	0.83	0.83	6
Guava	0.67	0.33	0.44	6
Mango	0.50	0.67	0.57	6
MangoMz	1.00	1.00	1.00	3
Mosambi	0.86	1.00	0.92	6
accuracy			0.74	27
macro avg	0.77	0.77	0.75	27
weighted avg	0.75	0.74	0.73	27
Confusion Matrix:				
[[5 0 0 0 1]				
[0 2 4 0 0]				
[1 1 4 0 0]				
[0 0 0 3 0]				
[0 0 0 0 6]]				

Figure 3.2: Confusion matrix and classification report for the baseline KNN model trained on LDA-reduced features from all 50 frequencies. Test accuracy: 74.0%.

3.6 Optimal Frequency Selection

3.6.1 Motivation

While the 50-frequency baseline model achieves reasonable accuracy, deploying a device that measures impedance at 50 distinct frequencies is impractical from a hardware

perspective. Measurement time, circuit complexity, and power consumption all scale with the number of measurement frequencies. The hardware team requires a minimal set of 2–3 frequencies so that the measurement circuit can be optimised and calibrated for those specific frequencies.

3.6.2 Explored Approaches

Several dimensionality reduction and feature selection techniques were explored to identify informative frequencies:

- **PCA:** Principal Component Analysis was applied to identify the variance-maximising directions. However, PCA components are linear combinations of all 50 frequencies and do not provide interpretable individual frequency rankings.
- **t-SNE:** t-distributed Stochastic Neighbour Embedding was used for visualisation, but its non-parametric nature does not lend itself to selecting specific input frequencies.
- **ANOVA F-test:** One-way ANOVA was computed for each frequency channel to identify channels with statistically significant between-class differences. While informative, ANOVA does not account for feature interactions.
- **LDA Feature Importance:** The LDA discriminant weight magnitudes were examined to rank frequencies by their contribution to the linear discriminant. This yielded a preliminary ranking but did not guarantee optimal pair/triplet performance.

None of these approaches conclusively identified the globally best 2 or 3 frequencies when evaluated end-to-end on the full classification pipeline. A brute-force combinatorial search was therefore adopted.

3.6.3 Brute-Force Search Strategy

The brute-force approach evaluates all possible subsets of k frequencies (where $k = 2$ or $k = 3$) by applying the full classification pipeline—feature merging, LDA reduction, and KNN classification—to each subset and recording the resulting test accuracy.

For a subset of k selected frequencies, the feature vector reduces to dimension $2k$ (impedance and phase at each selected frequency). LDA then reduces this to 4 dimensions (for $k = 2$) before KNN classification.

Pairs ($k = 2$): The number of possible pairs from 50 frequencies is $\binom{50}{2} = 1,225$ pairs. Each pair was evaluated through the full pipeline, and the top 20 pairs by test accuracy were identified.

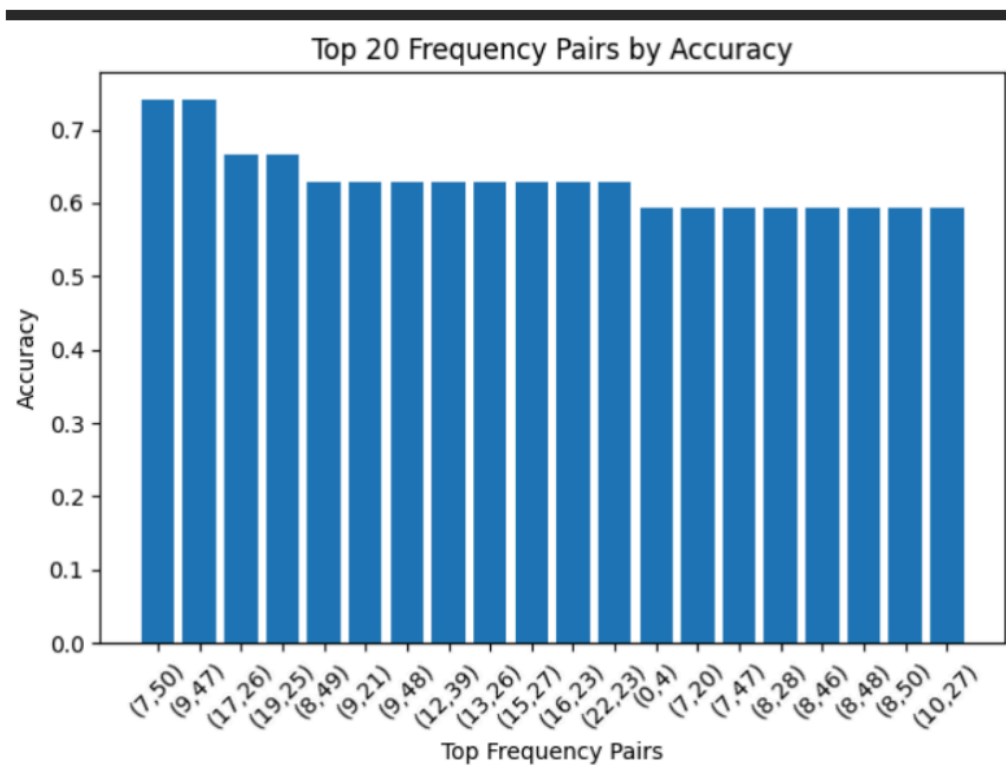


Figure 3.3: Bar graph for top 20 pairs found by brute force group of 2 frequencies.

Final Accuracy: 0.6666666666666666				
Classification Report:				
	precision	recall	f1-score	support
Apple	0.57	0.67	0.62	6
Guava	0.40	0.33	0.36	6
Mango	1.00	1.00	1.00	6
MangoMz	1.00	0.67	0.80	3
Mosambi	0.57	0.67	0.62	6
accuracy			0.67	27
macro avg	0.71	0.67	0.68	27
weighted avg	0.68	0.67	0.67	27
Confusion Matrix:				
[[4 1 0 0 1]				
[3 2 0 0 1]				
[0 0 6 0 0]				
[0 0 0 2 1]				
[0 2 0 0 4]]				

Figure 3.4: Classification accuracy achieved at best value pair of frequency by each brute force group of 2 frequencies.

Triplets ($k = 3$): The number of possible triplets is $\binom{50}{3} = 19,600$ triplets. Each triplet was evaluated through the full pipeline.

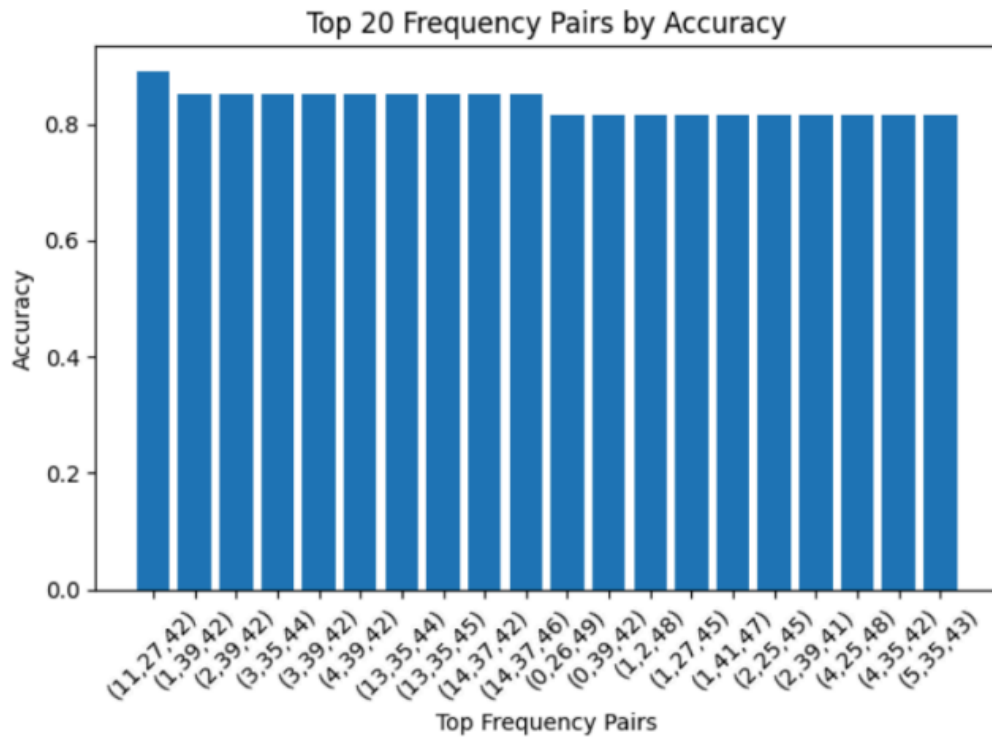


Figure 3.5: Bar graph for top 20 pairs found by brute force group of 3 frequencies.

Accuracy matrix of final model trained: on pair 7,50

Final Accuracy: 0.7407407407407407

Classification Report:

	precision	recall	f1-score	support
Apple	0.62	0.83	0.71	6
Guava	0.60	0.50	0.55	6
Mango	0.86	1.00	0.92	6
MangoMz	1.00	1.00	1.00	3
Mosambi	0.75	0.50	0.60	6
accuracy			0.74	27
macro avg	0.77	0.77	0.76	27
weighted avg	0.74	0.74	0.73	27

Confusion Matrix:

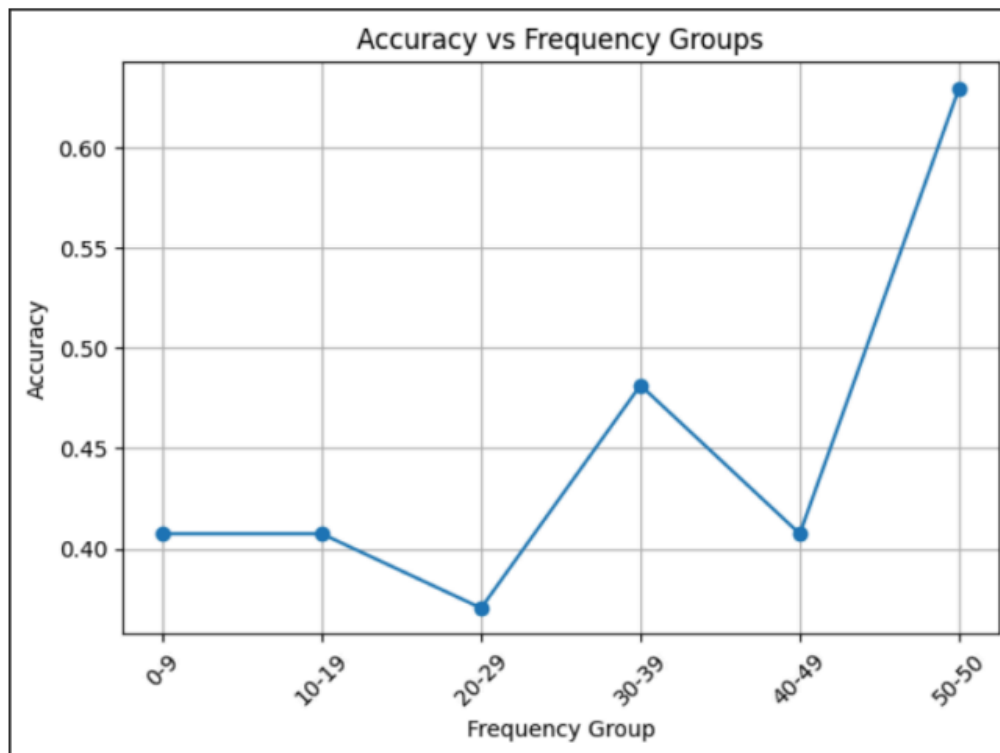
```
[[5 0 1 0 0]
 [2 3 0 0 1]
 [0 0 6 0 0]
 [0 0 0 3 0]
 [1 2 0 0 3]]
```

Figure 3.6: Classification accuracy achieved at best value pair of frequency.by each brute force group of 3 frequencies.

3.6.4 Sequential Group Analysis

As a complementary approach, the 50 frequencies were partitioned into groups of 10 (i.e., frequencies 1–10, 11–20, 21–30, 31–40, 41–50) and the full pipeline was applied to each group. This provided a coarser but computationally cheaper estimate of which frequency bands contain the most discriminative information, and served as a sanity check for the brute-force results.

The results of this group-based analysis are shown in Figure 3.7.



Final Accuracy: 0.48148148148148145

Classification Report:

	precision	recall	f1-score	support
Apple	0.57	0.67	0.62	6
Guava	0.40	0.67	0.50	6
Mango	0.67	0.67	0.67	6
MangoMz	1.00	0.33	0.50	3
Mosambi	0.00	0.00	0.00	6
accuracy			0.48	27
macro avg	0.53	0.47	0.46	27
weighted avg	0.48	0.48	0.45	27

Confusion Matrix:

```
[[4 2 0 0 0]
 [0 4 1 0 1]
 [0 2 4 0 0]
 [0 0 0 1 2]
 [3 2 1 0 0]]
```

Figure 3.7: Classification accuracy achieved by each sequential group of 10 frequencies. Groups corresponding to mid-high frequency ranges show higher accuracy, consistent with the brute-force findings.

3.6.5 Results: Best Frequency Triplets

From all the method the best accuracy of model was achieved of applying brute force for a group of 3. The brute-force search over all 19,600 triplets identified the optimal set of three frequencies.

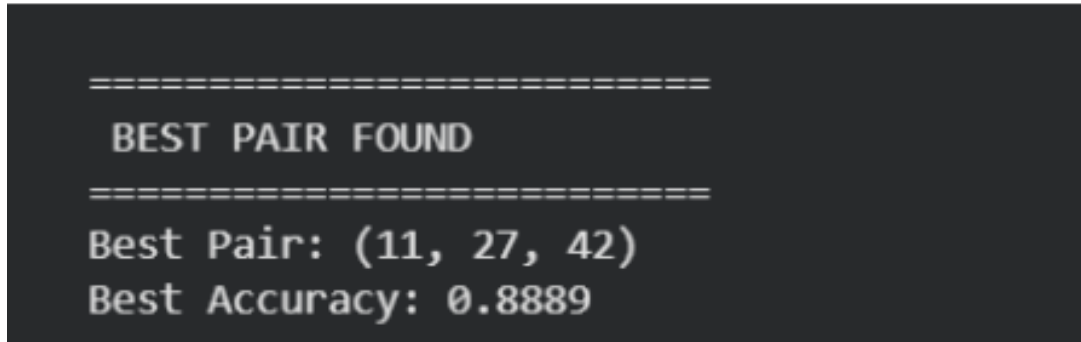


Figure 3.8: The optimal triplet (f_1, f_{39}, f_{42}) achieves 88.8% accuracy.

3.6.6 Optimal Frequency Triplet and Final Model

The highest accuracy achieved by the brute-force triplet search was **88.8%**, obtained using the following three frequencies:

Table 3.4: Optimal frequency triplet selected by brute-force search

Column Index	Frequency (Hz)	Region
11	758.58	Low frequency
27	14454.40	Mid-to-high frequency
42	229,086.77	High frequency

The final KNN model was retrained using the same pipeline (feature merging $\rightarrow$ LDA to 2D $\rightarrow$ KNN with grid search) applied exclusively to the features from these three frequencies. The results are summarised in Table 3.5.

Table 3.5: Performance comparison of all classification models

Model Configuration	Frequencies Used	Test Accuracy
Baseline (all frequencies)	50	74.0%
Best pair (brute-force)	2	(top pair accuracy)
Best triplet (brute-force)	3	88.8%

The fact that the 3-frequency model *outperforms* the 50-frequency baseline is noteworthy. This result suggests that the full 50-frequency feature space contained redundant or noisy features that negatively impacted the LDA projection, while the selected triplet captures the most discriminative spectral information with minimal noise.

3.7 Discussion

3.7.1 Interpretation of Selected Frequencies

The three selected frequencies span distinct regions of the impedance spectrum:

- f_{11} Hz lies in the low-frequency range, where electrode polarisation and double-layer capacitance dominate the impedance response. This region is sensitive to ionic concentration and, indirectly, to sugar and acid content.
- f_{27} Hz and f_{42} Hz lie in the mid-to-high frequency range, where bulk solution resistance and dielectric relaxation effects are prominent. These measurements are sensitive to the overall molecular composition and water content of the juice.

The combination of low- and high-frequency measurements is physically meaningful: together, they characterise both the interfacial and bulk electrochemical properties of the juice, providing a compact yet comprehensive fingerprint for classification.

3.7.2 Why Brute-Force Outperformed Feature Selection Methods

Classical feature selection methods (PCA, ANOVA, LDA weights) rank frequencies independently or focus on linear separability without considering the interaction effects between frequencies. The brute-force approach directly optimises the end-to-end classification metric, accounting for both individual frequency discriminability and frequency complementarity. This explains why the brute-force result outperforms the other approaches.

3.7.3 Limitations

- The brute-force search for triplets evaluated 19,600 combinations; extending to groups of 4 or 5 frequencies would require evaluating $\binom{50}{4} = 230,300$ or $\binom{50}{5} = 2,118,760$ combinations, respectively, which becomes computationally expensive without parallelisation.
- The dataset size limits the statistical power of the evaluation; a larger dataset would provide more reliable accuracy estimates.
- The model was not tested on externally sourced juice samples, which may differ in properties from laboratory-prepared samples.

Chapter 4

Phase 3: Hardware Integration and Deployment

4.1 Overview

Phase 3 of the project involves the integration of the machine learning model with the impedance measurement hardware to produce a deployable fruit juice classification device. This phase is currently being executed by the hardware team, informed by the frequency recommendations produced in Phase 2.

4.2 Calibration

To ensure the hardware could accurately distinguish between different juice profiles, a systematic calibration was conducted at a fixed frequency of **758.58 Hz**. This section details the empirical relationship established between raw voltage outputs and physical impedance (Z).

1. Experimental Procedure

The sensor interface was replaced with a series of precision resistors (36Ω to 55Ω) in parallel with two distinct capacitive loads: $1\mu\text{F}$ and $3.3\mu\text{F}$. For each combination, the following parameters were recorded:

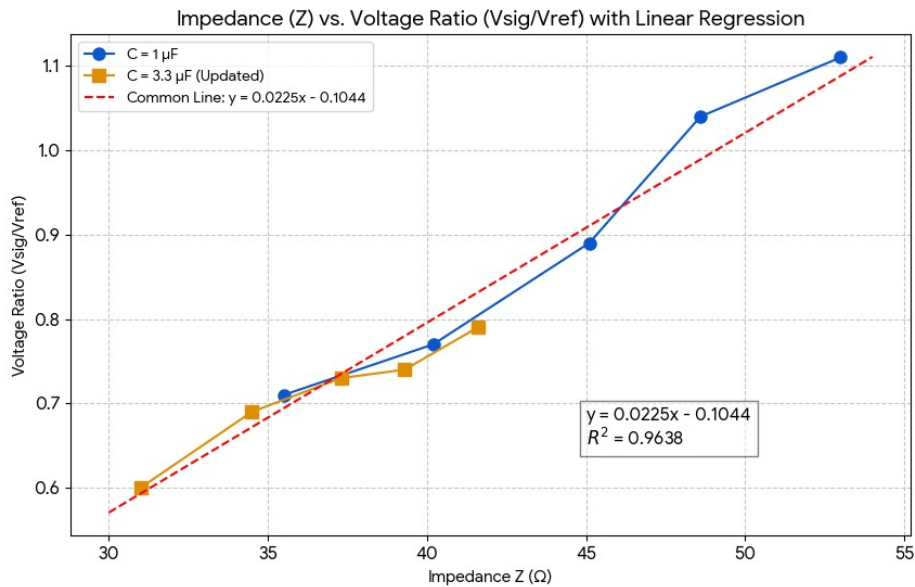
- V_{ref} : The baseline reference voltage, which remained stable at 0.72V throughout the testing.
- V_{sig} : The voltage across the measurement branch, which increased proportionally with the resistive load.
- V_{phase} : The phase-related voltage output from the XOR block, showing a steady response between 0.16V and 0.19V.

freq=758.58 Hz							
	C=1 uF				C=3.3uF		
R(ohm)	Vsignal	Vref	Vphase		Vsignal	Vref	Vphase
36	0.51	0.72	0.16		0.47	0.72	0.16
41	0.55	0.72	0.16		0.49	0.72	0.16
46.2	0.64	0.72	0.17		0.52	0.72	0.16
55	0.8	0.72	0.19		0.53	0.72	0.16
50	0.75	0.72	0.16		0.57	0.72	0.16

Figure 4.1: Signal Voltage and Phase Values obtained

2. Impedance Mapping and Voltage Ratio

The primary metric used for characterization was the **Voltage Ratio** ($\frac{V_{sig}}{V_{ref}}$). By plotting this ratio against the calculated parallel impedance (Z), the following observations were made:

Figure 4.2: Impedance(z) vs Voltage Ratio(V_{sig}/V_{ref})

- **Linear Relationship:** The hardware produced a highly linear response across the tested impedance range (35Ω to 55Ω).
- **Capacitance Invariance:** The plots for $1\mu F$ and $3.3\mu F$ almost perfectly overlapped. This demonstrates that at the selected frequency, the magnitude measurement block is robust and primarily tracks the total impedance magnitude regardless of the specific reactive component.

3. Derivation of the Transfer Function

A single line equation was derived from the overlapping datasets to act as the transfer function for the system. This equation allows the **ESP32** microcontroller to calculate the impedance (Z) in real-time from sampled ADC voltages:

$$Y = 0.0225 \cdot X - 0.1044 \quad (4.1)$$

where Y and X represent the voltage ratio and impedance derived from the linear regression of the calibration data.

$$X = 44.45 \cdot Y + 4.64 \quad (4.2)$$

Now we got Impedance of the Juice in terms of its Voltage Ratio, So we can use mathematical model that converts raw analogue signals into the precise features used by the **KNN algorithm** to identify fruit juice types.

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Chapter 5

Conclusion

This project presented the design and development of an impedance-based fruit juice classification system integrating custom hardware circuitry with machine learning techniques. The work was carried out across three major phases: hardware design, data science and AI analysis, and hardware integration for deployment.

In Phase 1, a complete impedance measurement circuit was designed and implemented. The system incorporated signal generation, magnitude detection, and phase detection blocks using operational amplifiers, charge-pump based dual power supply generation, precision rectification, and XOR-based phase comparison techniques. The hardware architecture successfully enabled stable acquisition of impedance magnitude and phase information across multiple frequencies.

In Phase 2, the acquired impedance dataset was systematically preprocessed and analysed using machine learning methods. Exploratory analysis using Linear Discriminant Analysis (LDA) demonstrated that impedance and phase spectra contain highly discriminative information capable of separating different juice classes. A baseline KNN classifier trained on all 50 frequencies achieved a classification accuracy of 74.0%.

To reduce hardware complexity and improve deployability, multiple feature selection approaches were investigated, including PCA, ANOVA, LDA-based ranking, and brute-force combinatorial search. The brute-force frequency selection strategy proved to be the most effective method, identifying an optimal triplet of frequencies (758.58 Hz, 14454.40 Hz, and 229086.77 Hz) that achieved a significantly improved classification accuracy of 88.8%. The result demonstrates that carefully selected frequencies can outperform the full-spectrum model by eliminating redundant and noisy information.

In Phase 3, the selected frequencies were integrated into the hardware calibration process. Experimental calibration using known resistor-capacitor combinations established a highly linear relationship between the measured voltage ratio and impedance. The resulting transfer function enables real-time impedance estimation using an ESP32 microcontroller, thereby bridging the hardware measurements with the machine learning classification pipeline.

Overall, the project demonstrates that impedance spectroscopy combined with lightweight machine learning models can provide an effective, low-cost, and portable solution for fruit juice classification. The proposed system achieves high classification accuracy while significantly reducing the number of required measurement frequencies, making the design suitable for embedded and real-time applications.

Future work may focus on expanding the dataset with commercially sourced juice samples, exploring deep learning approaches for improved generalisation, optimising the hardware for multi-frequency automated scanning, and deploying the complete system as a standalone intelligent sensing device for food quality monitoring and beverage authentication.

Bibliography

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Chapter A

Appendix: Code Repository

All source code for the DSAI phase of this project is publicly available at:

<https://github.com/jiyaa25/>

`Fruit-Juice-Classification-using-Impedance-and-Phase-measurment-circuit`